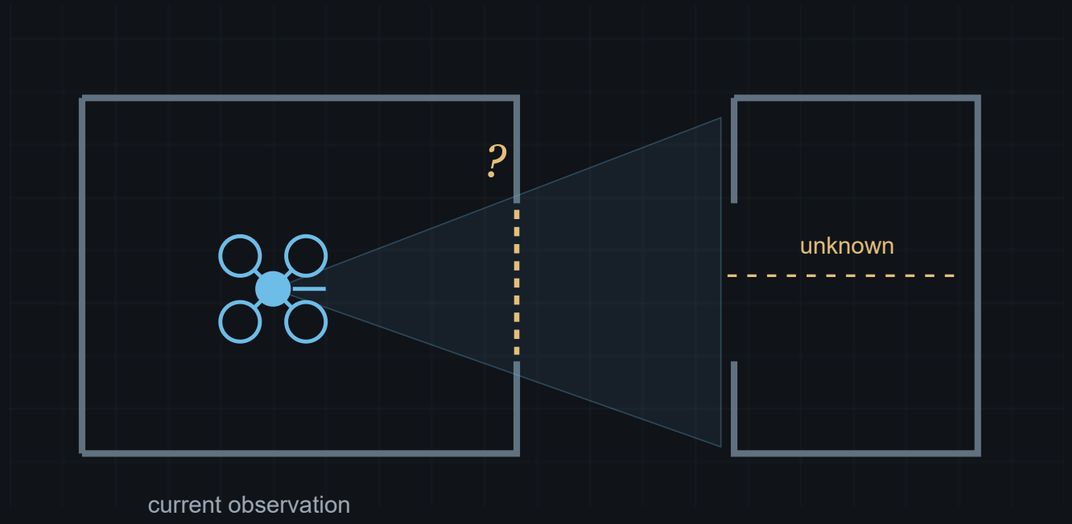
The course is visible.
The next action is uncertain.

Indoor navigation, incomplete observations, and a mission beyond the field of view.



What should happen next?

A passage candidate

What lies beyond it?

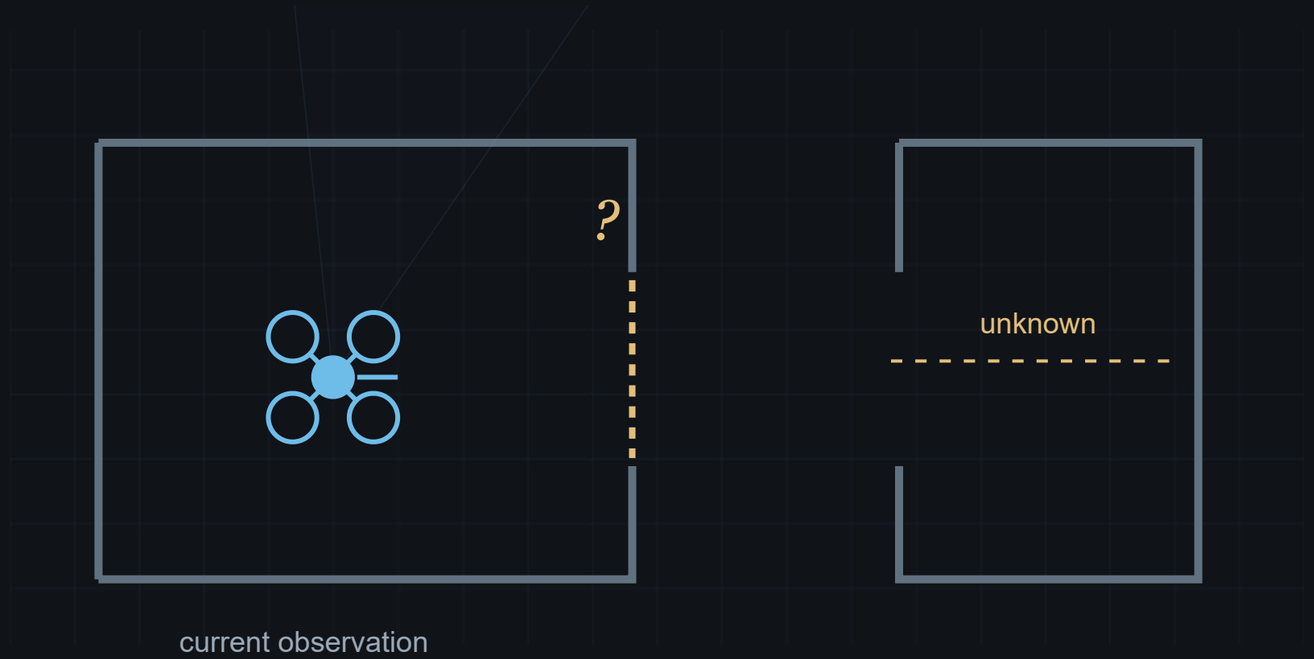
Observe before committing.

geometry + memory + intent

↳ Each camera frame reveals only part of the task.

Build what the drone *remembers*.

Remember openings, keep uncertainty, and look again when needed.



Memory ≠ a fresh view

opening_0003

same entity, new viewpoint

last seen → aging

identity persists · certainty does not

↳ Prioritize a useful world representation before increasing policy complexity.

Illustrative belief update

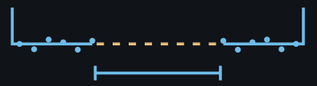
Give the map meaning.

Research gave us the structure; our constraints shaped the implementation.

RESEARCH → ADAPTATION

Hydra

build the map as we move



CSIRO: recovery in partial observation

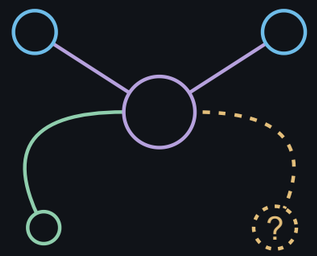
Kimera

connect geometry to meaning



ModalAI: separate perception, planning and autopilot interfaces

A useful world memory



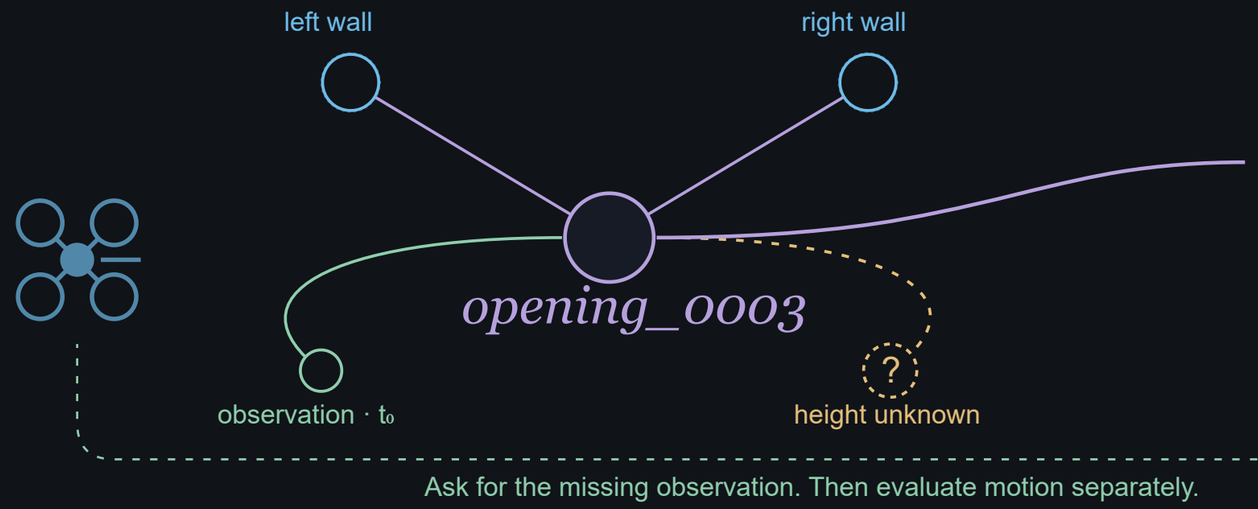
geometry + idopening + observations

↳ A compact adaptation for our task and compute budget.

From seeing to *understanding*.

Depth → walls and openings → WorldState → what to inspect next.

01 DEPTH OBSERVATIONS | 02 WALLS + OPENING | 03 WORLDSTATE | 04 REASONING



GOAL PROPOSAL

observe_more

target: opening_0003

because: height is unknown

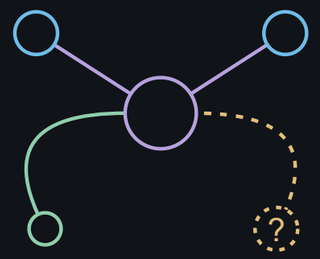
Grounded proposal ≠ permission to fly

↳ WorldState remembers what we saw—and what we still do not know.

Illustrative entity · offline contract

Sometimes the next action is *to look*.

If we do not know whether the drone fits, take another look.



opening_0003

height remains unknown

Read the memory

opening + observations

what we know + what is missing

do not assume a way through

Choose a useful action

observe_more

target: opening_0003

because: height is unknown

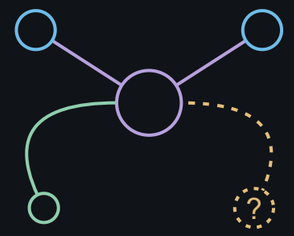
Check the evidence is current

Motion still needs its own feasibility check.

↳ When important information is missing, look again before committing.

Remember. Decide. Fly.

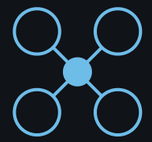
Three jobs, with a safety check before movement.











1. Remember

What is here?

2. Decide

What should we do next?

3. Fly safely

Can we make this move?



New observations update memory

↳ Reasoning chooses the goal. Control turns it into movement.

Why DATT and physics-informed learning?

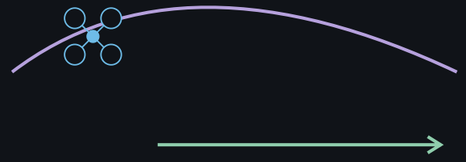
A good route still needs a drone that can follow it when conditions change.



Planned ≠ actual

wind · delay · model error

DATT · follow the route



Look ahead + adapt

Correct for disturbances

Proposed tracking experiment

PINNs · predict motion

Measurements

+ physical equations

A physics-guided prediction

Learn without ignoring physics

Our predictor: offline-tested

↳ Use learning to handle motion mismatch; keep world understanding separate.

Our Classical baseline: plan, then follow.

Established algorithms give us a clear starting point to measure against.

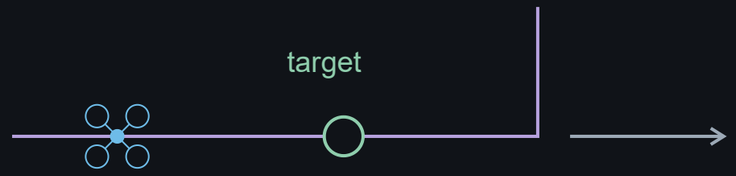
Plan a clear route



Dijkstra grid search

Live planner: A* alternative

Follow a sliding target



Look ahead; wait at corners

Limit lead to avoid cutting turns

Send position targets



PX4 flight control

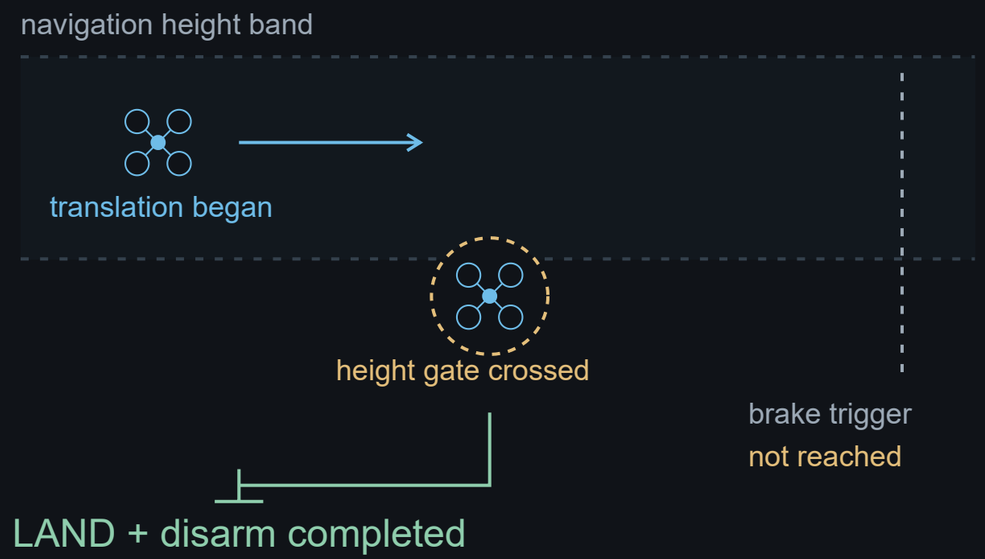
Monitor height and contact

↳ An understandable baseline lets us measure whether learning adds value.

What failed — and what we changed.

Simulation exposed problems with height, coordinates and timing.

HEIGHT DROPPED BEFORE THE BRAKING TEST



Schematic states and event order · not a measured trajectory

FOUR ATTEMPTS COULD NOT START



01 Coordinates disagreed

use the same position reference



02 Clock alignment

preserve capture-time identity



03–04 Data arrived too slowly

keep the command loop responsive

No course-flight success claimed.

↳ Fix height, shared coordinates and timing before adding more intelligence.

What we built. What we still need to prove.

Evidence is specific to a component, an experiment and an environment.

EVIDENCE IS LOCAL TO THE CLAIM



Offline components

entities · reasoning · mapping

synthetic comparison evidence



Simulation checks

Hover check: completed

Braking test: landed; test failed



Mission-level benefit

hybrid course + matched trials

hardware qualification remains

Separate baseline: Classical S1 has one successful Gazebo course run.

↳ The hybrid architecture has not yet demonstrated full-course native success in the fetched record.

From observations to *justified actions*.

The next milestone is a measurable mission benefit from each layer.



01 Fly one reliable segment

coordinates · timing · height



02 Matched comparison

same map, start and scoring



03 Layer ablations

memory → topology → semantics

Observations ⟶ *Grounded belief* ⟶ *Justified actions*

Learned extensions only when the evidence warrants them.

↳ Prove reliable flight, then measure what memory and reasoning add.

N1 · N6 · N7 · N8

Planned evaluation